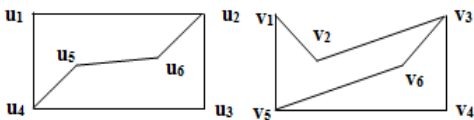
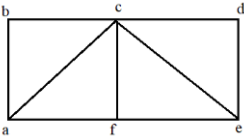
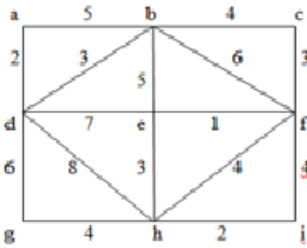


Course Code: B23BS2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
DISCRETE MATHEMATICS AND GRAPH THEORY					
(Common to CSE, AIML, CSBS, IT, AIDS, CSG, CIC, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Write the inverse, converse and contra positive of “ If $\triangle ABC$ is a right angle triangle then $AC^2 = AB^2 + BC^2$ ”	1	2	2
	b).	Translate the following statement into symbolic form. “Any integer is either positive or negative”.	1	2	2
	c).	Define relation matrix and give an example.	2	1	2
	d).	Define Lattice and give an example.	2	1	2
	e).	How many 4 digit numbers divisible by 5 can be formed using the digits 3, 7, 1, 5, 6.	3	2	2
	f).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$.	3	3	2
	g).	Define adjacency matrix.	4	1	2
	h).	Define Eulerian graph.	4	1	2
	i).	Define tree and give an example.	5	1	2
	j).	Define Graph coloring.	5	1	2
					5 x 10 =50Marks
		UNIT-1			
2.	a).	Prove that $\{[(p \vee q) \rightarrow r] \wedge \neg p\} \rightarrow (q \rightarrow r)$ is a tautology.	1	3	5
	b).	Verify that the following argument is valid by using the rules of inference If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun. If Clifton lives in France, then he rides a bicycle. Either Clifton speaks French, or he drives a Datsun. Hence, Clifton rides a bicycle.	1	3	5
		OR			
3.	a).	Verify that the following argument is valid by using the rules of inference, quantifiers. Babies are illogical. Nobody is despised who can manage a crocodile.	1	3	5

		Illogical people are despised. Hence, babies cannot manage crocodiles.			
	b).	Find the PDNF and PCNF of $p \vee \neg q$	1	3	5
		UNIT-2			
4.	a).	Find the number of integers between 1 and 250 which are divisible by any of the integers 2, 3, 5 or 7.	2	3	5
	b).	Let R denote a relation on the set of ordered pairs of positive integers such that $(x, y)R(u, v)$ if and only if $xv = yu$. Then establish that 'R' is an equivalence relation.	2	3	5
		OR			
5.	a).	Define Hasse diagram and draw Hasse diagram of $(P(A), \subseteq)$, where $A = \{1, 2, 3\}$.	2	3	5
	b).	Define bijective and inverse function. Find the inverse of the function $f(x) = 5x + 2$.	2	3	5
		UNIT-3			
6.	a).	A cricket team of 11 is to be selected out of 14 players of whom 5 are bowlers. Find the number of ways in which this can be done so as to include atleast 3 bowlers.	3	3	5
	b).	i) Determine the term independent of x in the expansion of $(x^2 + \frac{1}{x})^{12}$ ii) Determine the coefficient of $x^5 y^{10} z^5 w^5$ in the expansion $(x + 7y + 3z + w)^{25}$	3	3	5
		OR			
7.	a).	How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$.	3	3	5
	b).	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$ using generating functions.	3	3	5
		UNIT-4			
8.		Define isomorphism of graphs and determine the Isomorphism between the following graphs. 	4	3	10
		OR			
9.	a).	Prove that in any graph, there is an even number of vertices of odd degree.	4	3	5
	b).	Define Hamiltonian graph and find the Hamiltonian path and	4	3	5

		Hamiltonian graph in the following graph. 			
		UNIT-5			
10.	a).	State and Prove Euler's formula for planar graphs.	5	3	5
	b).	Show that a tree with n vertices has exactly (n-1) vertices.	5	3	5
		OR			
11.		Find the minimal spanning tree for the following weighted graph 	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23HS2101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
UNIVERSAL HUMAN VALUES-2: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT					
(Common to all programmes of Engineering)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the basic guidelines for value education?	1	2	2
	b).	What is MBTI personality test?	1	2	2
	c).	How can we differentiate between the needs of the Self and the needs of the Body?	2	2	2
	d).	What are the characteristics and activities of the Self (I)?	2	2	2
	e).	How is 'respect' defined in the context of human interaction?	3	2	2
	f).	How is society described in relation to the family?	3	2	2
	g).	How are the four orders of nature interconnected?	4	2	2
	h).	How does the idea of self-regulation in nature contribute to its harmony?	4	2	2
	i).	Define definitiveness of (ethical) human conduct.	5	2	2
	j).	Explain how humanistic education can influence professional ethics.	5	2	2
5 x 10 = 50 Marks					
		UNIT - I	CO	KL	M
2.	a).	Discuss natural acceptance	1	2	5
	b).	Differentiate prosperity and deprivation	1	2	5
		OR			
3.	a).	Deliberate the right understanding in perspective to self exploration.	1	2	5
	b).	What are the key functions of the MBTI? Explain.	1	2	5
		UNIT - II			
4.	a).	Illustrate coexistence of "I" and "Body ".	1	2	5
	b).	Distinguishing between the Needs of the Self and the Body	1	2	5
		OR			
5.	a).	Discuss Characteristic activities of Harmony with "I".	1	2	5
	b).	Explain Sanyam and Health.	1	2	5

		UNIT - III			
6.	a).	Write a note on human-human relationship as regarding harmony.	2	2	5
	b).	Differentiate intention and competence.	2	2	5
		OR			
7.	a).	Discuss salient values in relationship.	3	2	5
	b).	Illustrate universal Harmonious Society - an Undivided society.	3	2	5
		UNIT - IV			
8.		Discuss orders of life in nature and its significance self regulation of individual	4	2	10
		OR			
9.		Illustrate existence of human being as coexistence with universe in perspective of space	4	2	10
		UNIT - V			
10.		Discuss importance of professional competence for augmenting universal human order.	5	2	10
		OR			
11.	a).	Case study of typical holistic technologies.	5	2	5
	b).	Role of engineer in promoting harmony in society	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

II B.Tech., I Semester MODEL QUESTION PAPER

DIGITAL LOGIC & COMPUTER ORGANIZATION

Common to CSE, CSD, CIC & CSIT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Solve the following Boolean function to a minimum number of literals: $xy + x'z + yz$	1	3	2
	b).	Develop a 2X4 line Decoder.	1	3	2
	c).	Explain various types of buses?	2	2	2
	d).	Differentiate multi computers and multi processors.	2	2	2
	e).	Explain Instruction Format.	3	2	2
	f).	Describe zero address instruction Format with example.	3	2	2
	g).	Explain memory hierarchy.	4	2	2
	h).	Compare static RAM and dynamic RAM	4	2	2
	i).	Explain the need of I/O interface module.	5	2	2
	j).	Summarize various types of output peripheral devices.	5	2	2

Estd. 1980

AUTONOMOUS

5 x 10 = 50 Marks

		UNIT-1			
2.		Solve the Boolean Function $F(w, x, y, z) = \sum (1, 3, 7, 11, 15)$ into SOP and POS using K-Map.	1	3	10
		OR			
3.		Develop a Combinational Circuit for 8X1 line Multiplexer.	1	3	10
		UNIT-2			
4.		Construct an Arithmetic Logic Shift Unit.	2	3	10
		OR			
5.		Determine Booth Multiplication algorithm with example.	2	3	10
		UNIT-3			
6.	a).	Explain the organization of registers.	3	2	5
	b).	Describe about the Address Sequencing.	3	2	5
		OR			
7.		Illustrate different addressing modes with example.	3	2	10

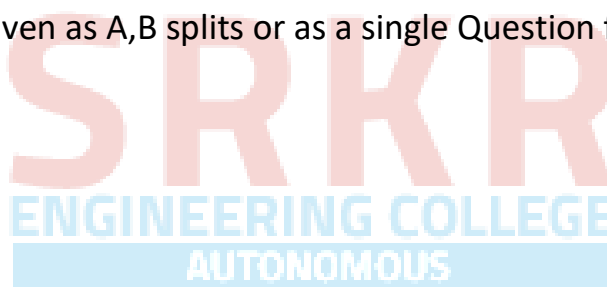
		UNIT-4			
8.	a).	Explain the hardware organization of associative memory with a neat block diagram.	4	2	5
	b).	Compare the relation between address and memory space in a virtual memory systems.	4	2	5
		OR			
9.		Explain about the mapping procedures of cache memory.	4	2	10
		UNIT-5			
10.	a).	Explain the functions of typical input-output interface.	5	2	5
	b).	Explain about DMA.	5	2	5
		OR			
11.		Describe the following with respect to asynchronous data transfer. a) Strobe control b) Handshaking c) Asynchronous serial transfer d) Asynchronous communication Interface.	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



Course Code: B23CS2102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
DATABASE MANAGEMENT SYSTEMS					
Common to CSE, CSG, CSIT & AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Compare DBMS and database.	1	2	2
	b)	Describe briefly two applications of DBMS	1	2	2
	c)	Describe an Entity set in ER model	2	2	2
	d)	Apply relational algebra selection operation for an example query	2	3	2
	e)	Illustrate a nested query in SQL	3	2	2
	f)	Use natural join in SQL to display data in two tables: t1 and t2	3	3	2
	g)	Explain normalization in RDBMS	4	2	2
	h)	Find whether the “E” attribute is key or not for relation R with F = {A->B, B->C, C->D, E->A}?	4	3	2
	i)	Describe Atomicity in brief	5	2	2
	j)	Discuss in brief about the log	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Summarize the differences between Database Management System and FPS	1	2	10
		OR			
3.		Illustrate Database system structure	1	2	10
		UNIT-2			
4.		Find different types of keys for the following tables Student(sid, sname, status), Catalog(sid,pid,cost) and Parts (pid, pname, color)	2	3	10
		OR			
5.		Apply conceptual DB design and draw ER diagrams for each situation for the University database containing information about departments (identified by DEPTNO), instructors (identified by SSN) and courses(identified by COURSEID). i. Each department can offer any number of courses	2	3	10

		ii. Many instructors can work in a department iii. An instructor can work only in one department iv. Each instructor can take any number of courses v. A course can be taken by only one instructor			
		UNIT-3			
6.		How do you use the different types of SQL statements (DDL and DML) on tables?	3	3	10
		OR			
7.		Apply different kinds of joins on your own example tables	3	3	10
		UNIT-4			
8.		How do you use functional dependencies to determine whether a table is in 2NF or 3NF?	4	3	10
		OR			
9.		Apply the steps to convert a table into BCNF using one example.	4	3	10
		UNIT-5			
10.		Illustrate the ARIES recovery algorithm with examples.	5	2	10
		OR			
11.		Describe different versions of 2PL protocols	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B23CS2103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. I Semester MODEL QUESTION PAPER					
OBJECT ORIENTED PROGRAMMING THROUGH JAVA					
Common to CSE, CSG, CIC, CSIT & AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
First Question is compulsory					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Compare POP and OOP?	1	2	2
	b).	Describe a scenario where using this keyword is necessary in a Java class.	1	2	2
	c).	What is type conversion in Java, and why is it necessary?	2	1	2
	d).	Explain String Constant Pool	2	2	2
	e).	What is Adapter class?	3	1	2
	f).	What are the uses of Super Keyword?	3	1	2
	g).	Explain about throw and throws keyword?	4	2	2
	h).	What are thread priorities in Java, and how do they affect the execution of threads?	4	1	2
	i).	What is File IO in Java?	5	1	2
	j).	List the steps to establish a database connection using JDBC	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain Features of JAVA.	1	2	5
	b).	Develop a program to accept two integers as command line arguments and print the sum of the two numbers.	1	3	5
		OR			
3.	a).	Explain Method Overloading with an example.	1	2	5
	b).	Explain different types of Constructors with suitable examples.	1	2	5
		UNIT-2			
4.	a).	Illustrate ArrayList and its methods.	2	2	5
	b).	Develop a program to reverse the elements of a given 2*2 array. Four integer numbers need to be passed as Command Line arguments.	2	3	5
		OR			
5.	a).	Illustrate HashMap and its methods.	2	2	5

	b).	Differentiate String and String Buffer class.	2	3	5
		UNIT-3			
6.	a).	Create a package called test package. Define a class called foundation inside the test package. Inside the class, you need to define 4 integer variables: var1 with private access modifier, var2 with default access modifier, var3 with protected access modifier and var4 with public access modifier. Import this class and packages in another class. Try to access all 4 variables of the foundation class and see what variables are accessible and what are not accessible.	3	3	5
	b).	Differentiate Interface and Abstract class.	3	3	5
		OR			
7.	a).	Explain different types of inheritance with an example.	3	2	5
	b).	Create an interface Vehicle with a default method message () that returns nothing and prints "Inside Vehicle". Create an interface FourWheeler with a default method message () that returns nothing and prints "Inside FourWheeler". Create a class Car implementing these two interfaces. In this class, Override the message () method and call the message () method of the Vehicle interface using super keyword. Instantiate the class, call the message method and print the output.	3	3	5
		UNIT-4			
8.	a).	Explain Multiple Catch Statements with an example.	4	2	5
	b).	Develop a java program to implement Thread Synchronization.	4	3	5
		OR			
9.	a).	Explain Thread Life Cycle with a neat diagram.	4	2	5
	b).	Develop a program that take an input String from user and parse it to integer, if it is not a number it will throw number format exception Catch it and print "Entered input is not a valid format for an integer." or else print the square of that number.	4	3	5
		UNIT-5			
10.	a).	Develop a program to copy contents of one file to another file.	5	3	5
	b).	Explain different types of JDBC drivers with neat diagrams.	5	2	5
		OR			
11.	a).	Develop a JDBC program to retrieve data from the database.	5	3	5
	b).	Explain the purpose of the Reader and Writer classes in Java. How do they differ from InputStream and OutputStream?	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B23HS2201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS					
(Common to AIDS, CSE, CIC, CSG, CSIT, CE, ECE, EEE, ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Define Managerial Economics.	1	1	2
	b)	State the Importance of Demand forecasting.	1	1	2
	c)	Write about Fixed cost and Variable cost.	2	1	2
	d)	List out the Applications of Break-even analysis.	2	1	2
	e)	Define Double Entry System of Accounting.	3	1	2
	f)	List the items under Current assets and Current liabilities.	4	1	2
	g)	Name the types of Imperfect Competition.	5	1	2
	h)	Identify the methods of Internet Pricing.	5	1	2
	i)	Show the components of working capital cycle.	6	1	2
	j)	Write the importance of Depreciation.	6	1	2
Estd. 1980 AUTONOMOUS					
5 x 10 =50Marks					
		UNIT-1	CO	KL	M
2.	a)	Compare the differences between Micro and Macro Economics.	1	2	5
	b)	Explain the Scope of Managerial Economics.	1	2	5
		OR			
3.	a)	Explain the determinants of Demand.	1	2	5
	b)	Describe the types of Elasticity of Demand.	1	2	5
		UNIT-2			
4.	a)	Illustrate the Elements of costs with suitable examples.	2	2	5
	b)	Define Cost. Explain the types of Costs.	2	2	5
		OR			
5.	a)	Interpret the determination of Break-even point with graphical representation.	2	2	5
	b)	Identify the Assumptions and Limitations of Break-even analysis.	2	2	5
		UNIT-3			

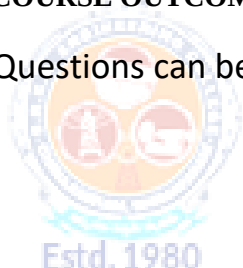
6.		Write the importance of Accounting and explain the types of accounts with rules governing each account.	3	2	10
		OR			
7.		Illustrate the proforma for Trading and Profit and loss account and Balance sheet including items in each account.	4	2	10
		UNIT-4			
8.	a)	Outline the salient features of Perfect competition.	5	2	5
	b)	Discuss the features of Oligopoly.	5	2	5
		OR			
9.	a)	Explain different methods of Cost Based Pricing.	5	2	5
	b)	Describe the Competition Based pricing methods.	5	2	5
		UNIT-5			
10.	a)	Discuss the factors influencing Working capital.	5	2	5
	b)	Explain the Sources of Raising finance in long term.	5	2	5
		OR			
11.	a)	Define Depreciation. Explain the causes of Depreciation in detail.	6	2	5
	b)	Explain the methods of Depreciation.	6	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

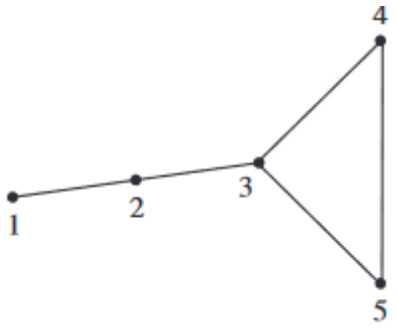
NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23BS2204																								
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)								R23																
II B.Tech. II Semester MODEL QUESTION PAPER																								
MATHEMATICAL MODELLING & SIMULATION																								
For CSG																								
Time: 3 Hrs.					Max. Marks: 70 M																			
Answer Question No.1 compulsorily																								
Answer ONE Question from EACH UNIT																								
Assume suitable data if necessary																								
								10 x 2 = 20Marks																
								CO	KL	M														
1.	a)	Define a mathematical model?						1	1	2														
	b)	Write the don'ts of the mathematical model?						1	1	2														
	c)	What is chebyshev approximation criterion?						2	1	2														
	d)	What is a residuals						2	2	2														
	e)	What are the methods of generating random numbers?						3	2	2														
	f)	What is a simulation						3	2	2														
	g)	Explain a transition matrix						4	2	2														
	h)	Define the reliability of system or component						4	1	2														
	i)	Illustrate Euler problem graphically?						5	2	2														
	j)	What is the use of vertex cover in integer programming?						5	2	2														
								5 x 10 =50Marks																
UNIT-1																								
2.	a)	Explain heuristic procedure to set up mathematical statements?						1	2	5														
	b)	Describe the classification of Mathematical models?						1	2	5														
OR																								
3.	a)	Explain the Modeling of a black box system						1	2	5														
	b)	Describe state variables and system parameters						1	2	5														
UNIT-2																								
4.		The following data represent the growth of a population of fruit flies over a 6-week period. Test the following models by plotting an appropriate set of data. Estimate the parameters of the following models. i) $P = ct$ ii) $P = a e^{bt}$ <table><tr><td>t (days)</td><td>7</td><td>14</td><td>21</td><td>28</td><td>35</td><td>42</td></tr><tr><td>P (the number of observed flies)</td><td>8</td><td>41</td><td>133</td><td>250</td><td>280</td><td>297</td></tr></table>						t (days)	7	14	21	28	35	42	P (the number of observed flies)	8	41	133	250	280	297	2	3	10
t (days)	7	14	21	28	35	42																		
P (the number of observed flies)	8	41	133	250	280	297																		
OR																								
5.		Derive the equations that minimize the sum of the squared deviations						2	3	10														

		between a set of data points and the quadratic model $y = c_1x^2 + c_2x + c_3$. Use the equations to find estimates of c_1, c_2 and c_3 for the following set of data. <table border="1"> <tr> <td>x</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td></tr> <tr> <td>y</td><td>0.06</td><td>0.12</td><td>0.36</td><td>0.65</td><td>0.95</td></tr> </table> And also Compute D , d_{max} to bound c_{max}	x	0.1	0.2	0.3	0.4	0.5	y	0.06	0.12	0.36	0.65	0.95			
x	0.1	0.2	0.3	0.4	0.5												
y	0.06	0.12	0.36	0.65	0.95												
		UNIT-3															
6.	a)	Use the middle-square method to generate 10 random numbers using $x_0 = 1009$.	3	3	5												
	b)	Explain the Monte-Carlo fair coin Algorithm	3	2	5												
		OR															
7.		Using the Monte - Carlo simulation find the area under curve and also develop the Monte-Carlo area algorithm	3	3	10												
		UNIT-4															
8.		Consider a more advanced stereo system with component reliabilities as displayed in Figure. Determine the system's reliability. What assumptions are required? <i>FIG: More advanced stereo system</i>	4	3	10												
		OR															
9.		Consider a model for the long-term dining behavior of the students at College SRKR. It is found that 25% of the students who eat at the college's Dining Hall-I return to eat there again, whereas those who eat at Dining Hall-II have a 93% return rate. These are the only two dining halls available on campus, and assume that all students eat at one of these halls. Formulate a model to solve for the long-term percentage of students eating at each hall.	4	3	10												

		UNIT-5																		
10	Consider the graph shown in Figure  i) Find a minimum vertex cover in the graph in Figure ii) Formulate an integer program to find a minimum vertex cover in the graph	5	3	10																
		OR																		
11	A basketball coach needs to find a starting lineup for her team. There are five positions that must be filled: point guard (1), shooting guard (2), swing (3), power forward (4), and center (5). Given the data in Table , create a graph model and use it to find a feasible starting lineup. What changes if the coach decides she can't play Latha in position 3? Table <table> <tr> <td>Devi</td> <td>Rani</td> <td>Manasa</td> <td>Laxmi</td> <td>Uma</td> <td>Rajiya</td> <td>Marry</td> <td>Latha</td> </tr> <tr> <td>1,2</td> <td>1</td> <td>1,2</td> <td>3,4,5</td> <td>2</td> <td>1</td> <td>3,4</td> <td>2,3</td> </tr> </table>	Devi	Rani	Manasa	Laxmi	Uma	Rajiya	Marry	Latha	1,2	1	1,2	3,4,5	2	1	3,4	2,3	5	3	10
Devi	Rani	Manasa	Laxmi	Uma	Rajiya	Marry	Latha													
1,2	1	1,2	3,4,5	2	1	3,4	2,3													

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B23CS2201																				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23															
II B.Tech. II Semester MODEL QUESTION PAPER																				
OPERATING SYSTEMS																				
Common to CSE,CSG, CSIT & CIC																				
Time: 3 Hrs.			Max. Marks: 70 M																	
Answer Question No.1 compulsorily																				
Answer ONE Question from EACH UNIT																				
Assume suitable data if necessary																				
10 x 2 = 20 Marks																				
			CO	KL	M															
1.	a).	List the services of Operating system?	1	1	2															
	b).	Differentiate fork() and vfork()	1	2	2															
	c).	What does PCB contain?	2	1	2															
	d).	Differentiate long term and short term scheduler?	2	2	2															
	e).	List the necessary conditions for a deadlock situation to arise?	3	1	2															
	f).	Define race condition	3	1	2															
	g).	List any two disadvantages of Thrashing?	4	1	2															
	h).	Differentiate Internal and external fragmentation.	4	2	2															
	i).	List different File Attributes	5	1	2															
	j).	List the different types of directory in OS	5	1	2															
5 x 10 = 50 Marks																				
UNIT-1																				
2.	a).	Explain Operating System Structures?	1	2	5															
	b).	Explain briefly different types of System calls?	1	2	5															
OR																				
3.		Explain the different functions of an operating system and discuss the various services provided by an operating system	1	2	10															
UNIT-2																				
4.	a).	What is a thread? Discuss about thread scheduling.	2	2	5															
	b).	Explain in detail Inter Process Communication?	2	2	5															
OR																				
5.		Evaluate preemptive and non-preemptive SJF CPU Scheduling algorithm for given Problem	2	3	10															
		<table><tr><td>Process</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td></tr><tr><td>Process Time</td><td>8</td><td>4</td><td>9</td><td>5</td></tr><tr><td>Arrival Time</td><td>0</td><td>1</td><td>2</td><td>3</td></tr></table>				Process	P1	P2	P3	P4	Process Time	8	4	9	5	Arrival Time	0	1	2	3
Process	P1	P2				P3	P4													
Process Time	8	4	9	5																
Arrival Time	0	1	2	3																

		UNIT-3			
6.	a).	Explain about Deadlock Avoidance?	3	2	5
	b).	Explain how semaphores are useful while for solving Dining-Philosophers Problem	3	2	5
		OR			
7.	a).	Explain Banker's Algorithm with an Example?	3	3	10
		UNIT-4			
8.	a).	What is virtual memory? Discuss the benefits of virtual memory techniques.	4	2	5
	b).	Write a short notes on Disk management.	4	2	5
		OR			
9.	a).	Consider the following reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Assume there are three frames. Apply LRU replacement algorithm to the reference sting above and find out how many page faults are produced. Illustrate the LRU page replacement algorithm in detail and also two feasible implementation of the LRU algorithm.	4	3	5
	b).	Explain the following disk scheduling algorithm with proper diagram a) FCFS b) LOOK c) C-SCAN.	4	2	5
		UNIT-5			
10.	a).	Explain file allocation methods in detail.	5	2	5
	b).	Discuss the advantages of maintain directories.	5	2	5
		OR			
11.	a).	Explain about access matrix and implementation of aces matrix.	5	2	5
	b).	Write short notes on a)Directory Implementation b)File system Structure	5	2	5

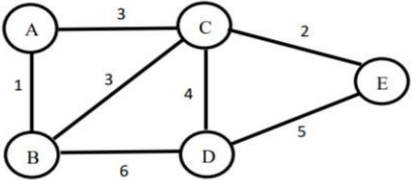
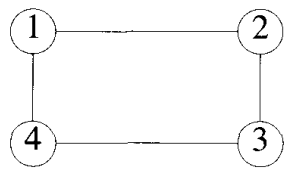
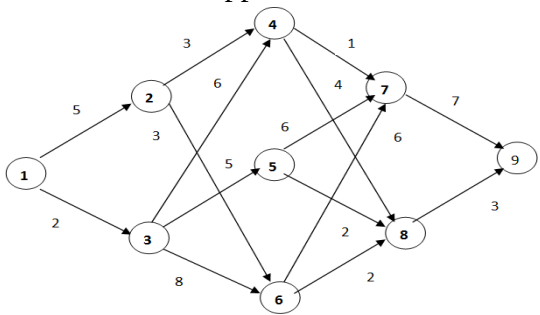
CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks

Course Code: B23CS2202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
ADVANCED DATA STRUCTURES AND ALGORITHM ANALYSIS					
Common to CSE,CSG, CSIT & AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	List out the properties of a Red-Black Tree	1	2	2
	b).	How to check Bi-connectivity in a graph?	1	2	2
	c).	Describe Asymptotic notations.	2	2	2
	d).	Explain the general method of divide and conquer.	2	2	2
	e).	How does the Greedy Method solve the Fractional Knapsack Problem?	3	2	2
	f).	Outline Backtracking strategy.	3	2	2
	g).	Why is Dynamic Programming more efficient than simple recursion for certain problems?	4	2	2
	h).	Explain the role of state and transition in dynamic programming with an example.	4	2	2
	i).	Discuss NP-Hard problems.	5	2	2
	j).	Describe Chromatic Number Decision Problem (CNDP).	5	2	2
					5 x 10 = 50 Marks
		UNIT-1			
2.		Explain AVL Tree rotations with examples	1	2	10
		OR			
3.	a).	Compare and contrast adjacency matrix and adjacency list representations of graphs.	1	2	5
	b).	Describe the Breadth-First Search (BFS) algorithm.	1	2	5
		UNIT-2			
4.		Describe the Quick Sort algorithm and analyze its average-case time complexity.	2	4	10
		OR			
5.		Explain the Divide and Conquer approach for finding the Convex Hull of a set of points and analyze its time complexity.	2	4	10
		UNIT-3			

6.	a).	Solve the following job sequencing with deadlines using greedy method. $n=6, (p_1, p_2, \dots, p_6) = (20, 40, 5, 15, 10, 8)$ and $(d_1, d_2, \dots, d_6) = (5, 2, 4, 3, 3, 1)$	3	3	5
	b).	Find the Minimum Spanning for the following graph using Kruskal's algorithm. 	3	3	5
		OR			
7.	a).	Explain Graph Coloring problem and Construct State Space Tree when $n=4$ and $m=3$. 	3	3	5
	b).	Solve 0/1 knapsack problem for the following data using Back Tracking. $n=5, (p_1, p_2, p_3, p_4, p_5) = (7, 8, 9, 11, 12), (w_1, w_2, w_3, w_4, w_5) = (13, 15, 16, 23, 24)$ with knapsack capacity, $m=26$.	3	3	5
		UNIT-4			
8.		Build OBST for given data $n=4, (a_1, a_2, a_3, a_4) = (\text{do}, \text{if}, \text{int}, \text{while})$ and $p(1:4) = (3, 3, 1, 1)$ $q(0:4) = (2, 3, 1, 1, 1)$.	4	3	10
		OR			
9.		Find a minimum cost path from S to T graph of following figure. Do this in forward Approach. 	4	3	10
		UNIT-5			
10.	a).	Consider the travelling sales person instance defined by the cost matrix. Find out minimum cost path using LC Branch-Bound.	5	3	5

		$\begin{bmatrix} \infty & 2 & 10 & 5 \\ 2 & \infty & 9 & \infty \\ 4 & 3 & \infty & 4 \\ 6 & 8 & 7 & \infty \end{bmatrix}$			
	b).	State and explain cook's theorem	5	2	5
		OR			
11.	a).	Draw the portion of the State Space Tree generated by LCBB for the instances: $n=5, m=12, (p1 \dots p5) = (10, 15, 6, 8, 4), (w1 \dots w5) = (4, 6, 3, 4, 2)$	5	3	5
	b).	Explain Classes NP-hard and NP-complete	5	2	5
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CD2201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
HUMAN COMPUTER INTERACTION					
Common to CSG & CSIT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Discuss the key characteristics of good and poor design?	1	2	2
	b).	Identify the primary goals of usability in the context of user experience design?	1	2	2
	c).	Define cognition in the context of human-computer interaction?	2	1	2
	d).	Identify and describe major cognitive frameworks used in interaction design?	2	2	2
	e).	Describe the relationship between user emotions and the success of a digital product?	3	2	2
	f).	Identify common characteristics of annoying interfaces?	3	2	2
	g).	List the key differences between quantitative and qualitative data analysis?	4	2	2
	h).	List the key principles of effective data visualization?	4	2	2
	i).	Define the interactive design process and its key components?	5	1	2
	j).	Define usability studies and their role in interactive design?	5	1	2
5 x 10 =50 Marks					
		UNIT-1			
2.	a).	Discuss the key characteristics that differentiate good design from poor design in interaction design?	1	2	5
	b).	Explain the relationship between interaction design and user experience. How do design decisions influence the user experience?	1	2	5
		OR			
3.	a).	What are conceptual models in interaction design, and why are they important?	1	2	5
	b).	Compare and contrast different models and frameworks used in interaction design.	1	2	5
		UNIT-2			
4.	a).	How do cognitive processes such as perception, attention, memory, and learning affect the way users interact with digital systems?	2	2	5

	b).	Compare different cognitive frameworks and discuss their relevance in designing intuitive and user-friendly interfaces.	2	3	5
		OR			
5.	a).	What strategies can designers employ to enhance social engagement through digital interfaces?	2	2	5
	b).	Evaluate the effectiveness of current technologies in achieving co-presence and the potential future developments?	2	3	5
		UNIT-3			
6.	a).	Explain how emotional interaction impacts user experience in digital interfaces?	3	2	5
	b).	Explain how emotional AI can enhance user interactions with digital interfaces?	3	2	5
		OR			
7.	a).	Explain how persuasive technologies can influence user behaviour?	3	2	5
	b).	Explain how natural user interfaces differ from traditional graphical user interfaces (GUIs)?	3	2	5
		UNIT-4			
8.	a).	Discuss the difference between data analysis, interpretation, and presentation, and explain how they interrelate?	4	2	5
	b).	Explain the importance of selecting appropriate data collection methods for different research objectives?	4	2	5
		OR			
9.	a).	Explain how data visualization techniques can enhance the understanding and communication of complex data?	4	2	5
	b).	Discuss the importance of ethical practices in data analysis, particularly in ensuring privacy, consent, and data integrity?	4	2	5
		UNIT-5			
10.	a).	Create a low-fidelity prototype for a proposed digital product and explain the decisions made during its development?	5	3	5
	b).	Develop a conceptual design for a new user interface, outlining the key concepts and user interactions?	5	3	5
		OR			
11.	a).	Design and conduct a usability study for a digital product, then interpret the results to suggest design improvements?	5	3	5
	b).	Design an experiment to test a specific aspect of a user interface, outlining the variables, controls, and expected outcomes?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 14 marks